

Dante Domizzi Sánchez Gallegos

Postdoc

ddomizzisag@gmail.com
Mexican

Social Network

ResearchGate

Github Projects

Languages

Spanish

English

Software

Virtual containers

Docker and Kubernetes

Microservices

Latex

C/C++

Java

Python

Education

2020 – 2024 **PhD on computer science and technology** — Universidad Carlos III de Madrid
Thesis: New techniques to build and manage agnostic workflows for the processing of digital products.
Advisor: Dr. Jesus Carretero.

2019 – 2023 **PhD on science in engineering and computational technologies** — Cinvestav Tamaulipas
Thesis: A continuity processing method for building and managing infrastructure-agnostic processing structures.
Advisor: Dr. Jesus Carretero.

2017 – 2019 **Master on science in engineering and computational technologies** — Cinvestav Tamaulipas
Thesis: Construction model of agnostic workflows to manage the life cycle of earth observation products.
Advisors: Dr. José Luis González Compeán and Dr. Victor Sosa Sosa.

2013 – 2016 **Engineering in IT** — Universidad Politécnica de Victoria
Degree project: Focused web crawler based on web pages and semantic resources classification.
Advisor: Dra. Heidy Marisol Marín Castro.

Professional experience

January 2024	Postdoc	Universidad Carlos III de Madrid
June 2023 – December 2024	Research Assistant	Universidad Carlos III de Madrid
February 2024 – April 2024	Software Tester - Pentesting	Cinvestav
February 2023 – April 2023	Software Tester	Cinvestav

Participation on research projects

2026 – 2027 **New techniques for managing data factories in the computing continuum**
Universidad Carlos III de Madrid — UC3M Programa Propio, Ref. 2025/00822/001
Principal Researcher

2024 – 2026 **New scalable I/O techniques for hybrid HPC and data-intensive workloads (SCIOT)**
Universidad Carlos III de Madrid — Agencia Española de Investigación

2023 – 2024 **DECIDE: New Methods on High-End and Edge Computing for the Intensive Processing of Data**
Universidad Carlos III de Madrid — Ministerio de ciencia, innovación y universidades

2020 – 2022 **ConvergenciA Big dAta-Hpc: de Los sensores a las Aplicaciones (CABAHLA-CM)**
Universidad Carlos III de Madrid — Comunidad De Madrid

2020 – 2023 **Technological platform for the management, assurance, exchange, and storage of large volumes of medical data, and cbuilding of a national repository of health data analytics services.**
Conacyt—PRONACES

2017 **Cloud computing federated service to store, process, and distribute satellite imagery**
Conacyt—AEM

Research stays

May 2026 – July 2026	**Scientific Computing and Imaging Institute, The University of Utah, USA** **Project**: Intelligent data storage systems for the computing continuum	Advisor: Prof. Manish Parashar
January 2024 – March 2024	**Globus Laboratory, University of Chicago** **Project**: DynoStore – A resilent data fabric system	Advisors: Kyle Chard and Ian Foster
October 2022 – February 2023	**CICESE Monterrey** **Project**: Arquitecturas de cómputo edge-fog-cloud	Advisor: Dr. Alejandro Galaviz
October 2018 – November 2018	**Parthenope University of Naples** **Project**: DagOnStar	Advisor: Dr. Raffaele Montella
January 2017 – August 2017	**Cinvestav Tamaulipas** **Project**: Cloud computing federated service to store, process, and distribute satellite imagery.	Advisor: Dr. José González

Publications

2025 **Nez: A design-driven skeleton model for building continuum AI-based and analytic systems.**
Sanchez-Gallegos, D. D., Carrizales-Espinoza, D., Gonzalez-Compean, J. L., Nunez-Gaona, M. A., Aguirre-Meneses, H., & Carretero, J.
Information and Software Technology, 107911.

2025 **Xook-Sec: A policy-as-code framework for secure data-sharing on the computing continuum.**
Torres-Charles, C. A., Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., Morales-Sandoval, M., & Carretero, J.
Cluster Computing, 28(14), 889.

2025 **Smart Cities: A Systematic Review of Emerging Technologies.**
Sanchez-Gallegos, D. D., Carrizales-Espinoza, D. E., Torres-Charles, C., & Carretero, J.
Smart Cities (2624-6511), 8(5).

2025 **OrbiFaaS: An orbital method to build Continuum Earth Observation Systems**
Torres-Charles, Catherine and Sanchez-Gallegos, Dante D and Carrizales-Espinoza, Diana and Gonzalez-Compean, JL & Carretero, Jesus
Future Generation Computer Systems, 174, 107992

2024 **StructMesh: A storage framework for serverless computing continuum**
Diana Carrizales-Espinoza, Dante D Sanchez-Gallegos, JL Gonzalez-Compean, & Jesus Carretero
Future Generation Computer Systems, 159, 353-369

2023 **A novel approach for large-scale environmental data partitioning on cloud and on-premises storage for compute continuum applications**
Mellone, G., De Vita, C. G., Sánchez-Gallegos, D. D., Sánchez-Gallegos, G., Torres-Charles, C. A., Garcia-Blas, J., ... & Laccetti, G.
Concurrency and Computation: Practice and Experience, 35(25), e7893.

2023 **On the building of efficient self-adaptable health data science services by using dynamic patterns.**
Sanchez-Gallegos, G., Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., Reyes-Anastacio, H. G., & Carretero, J.
Future Generation Computer Systems, 145, 478-495.

2023 **CD/CV: Blockchain-based schemes for continuous verifiability and traceability of IoT data for edge–fog–cloud**
Martinez-Rendon, C., González-Compeán, J. L., Sánchez-Gallegos, D. D., & Carretero, J.
Information Processing & Management, 60(1).

2022 **On the Efficient Delivery and Storage of IoT Data in Edge–Fog–Cloud Environments**
Barron, A., Sanchez-Gallegos, D. D., Carrizales-Espinoza, D., Gonzalez-Compean, J. L., & Morales-Sandoval, M.
Sensors, 22(18), 7016.

2022 **PuzzleMesh: A puzzle model to build mesh of agnostic services for edge-fog-cloud**
Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., Carretero, J., Marin, H., Tchernykh, A., & Montella, R.
IEEE Transactions on Services Computing.

2022 **FedFlow: A federated platform to build secure sharing and synchronization services for health dataflows**
Carrizales-Espinoza, D., Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., & Carretero, J.
Computing, 1-19.

2021 **An efficient pattern-based approach for workflow supporting large-scale science: The DagOnStar experience**
Sánchez-Gallegos, D. D., Di Luccio, D., Kosta, S., Gonzalez-Compean, J. L., & Montella, R.
uture Generation Computer Systems, 122, 187-203.

2020 **From the edge to the cloud: A continuous delivery and preparation model for processing big IoT data**
Sánchez-Gallegos, D. D., Carrizales-Espinoza, D., Reyes-Anastacio, H. G., Gonzalez-Compean, J. L., Carretero, J., Morales-Sandoval, M., & Galaviz-Mosqueda, A.
Simulation Modelling Practice and Theory, 105, 102136.

2020 **On the continuous processing of health data in edge-fog-cloud computing by using micro/nanoservice composition**
Sánchez-Gallegos, D. D., Galaviz-Mosqueda, A., Gonzalez-Compean, J. L., Villarreal-Reyes, S., Perez-Ramos, A. E., Carrizales-Espinoza, D., & Carretero, J.
IEEE Access, 8, 120255-120281.

Publications in conferences, congresses, and workshops

2026 **OsteoCAD: A Human-in-the-Loop Cloud-Edge Framework for Bone Tumor Segmentation**
Rodriguez-Herrero, Maximo and Sanchez-Gallegos, Dante D and Aguirre-Meneses, Heriberto and Núñez-Gaona, Marco Antonio and Gonzalez-Compean, JL and Carretero, Jesus
6th IEEE International Conference on ICT Solutions for eHealth

2026 **On modeling knowledge graphs for representing and explaining wide-area distributed storage system**
Morabito, Gabriele and Sanchez-Gallegos, Dante (presenter) and Fazio, Maria and Carretero, Jesus.
HPCMALL@SC/HPCASIA

2025 **A Computer-aided Framework for Detecting Osteosarcoma in Computed Tomography Scans**
Rodriguez-Herrero, Maximo and Sanchez-Gallegos, Dante (presenter) and Núñez-Gaona, Marco Antonio and Aguirre-Meneses, Heriberto and Gutiérrez, Luis Alberto Villalvazo and Velasco, Mario Ibrahin Gutiérrez and Gonzalez-Compean, JL and Carretero, Jesus.
HiPES@Europar

2025 **DAGonStore: Reliable Data Management for Workflows on the Computing Continuum with DynoStore and DAGonStar**
Sanchez-Gallegos, D. D. (presenter), Gonzalez-Compean, J. L., Carretero, J., & Montella, R.
WORKS@SC'25 2025

2025 **D-Rex: Heterogeneity-Aware Reliability Framework and Adaptive Algorithms for Distributed Storage**
Maxime Gonthier, Dante D Sanchez-Gallegos, Haochen Pan, Bogdan Nicolae, Sicheng Zhou, Hai Duc Nguyen, Valerie Hayot-Sasson, J Gregory Pauloski, Jesus Carretero, Kyle Chard, Ian Foster
ICS 2025

2025 **DynoStore: a wide-area distribution method for the dynamic allocation of data over heterogeneous storage**
Dante D. Sanchez-Gallegos, J. L. Gonzalez-Compean, Maxime Gonthier, Valerie Hayot-Sasson, J. Gregory Pauloski, Haochen Pan, Kyle Chard, Jesus Carretero and Ian Foster
CCGRID'25

2024 **Computing continuum systems using serverless abstractions**
Dante D. Sanchez-Gallegos (presenter), & Jesus Carretero.
Workshops of EuroPar Phd-Symposium

2024 **GeoNimbus: A serverless framework to build earth observation and environmental services**
Diana Carrizales-Espinoza, Alejandro Zequeira, Catherine Torres, J. L. Gonzalez-Compean, & Jesus Carretero.
Workshops of EuroPar HiPES 2024

2024 **An ad-hoc file system accelerated workflow application for accidental fire fast response**
Ciro Giuseppe De Vita, Dario Caramiello, Gennaro Mellone, Genaro Sánchez-Gallegos, Dante Domizzi Sánchez-Gallegos, Valeria Mele, Stefania Cavallo, Diana Di Luccio
2nd Workshop on Workflows in Distributed Environments

2023 **A data science pipeline synchronisation method for edge-fog-cloud continuum**
Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., Carretero, J., & Marin-Castro, H.
SC'23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis

2023 **eScience serverless data storage services in the edge-fog-cloud continuum**
Sanchez-Gallegos, D. D., Carrizales-Espinoza, D., Gonzalez-Compean, J. L., & Carretero, J.
2023 IEEE 19th International Conference on e-Science (e-Science) (pp. 1-9). IEEE.

2023 **Modelo de sincronización para la mitigación de cuellos de botella en flujos de datos**
Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., & Carretero, J.
Jornadas SARTECO 2023 (pp. 281-288)

2023 **Citizen Science for the Sea with Information Technologies: an open platform for gathering marine data and marine litter detection from leisure boat instruments**
De Vita, C. G., Mellone, G., Sanchez-Gallegos, D. D., Coviello, G., Romano, D., Lapegna, M., & Ciaramella, A
2023 IEEE 19th International Conference on e-Science (e-Science) (pp. 1-9). IEEE.

2023 **A containerized distributed processing platform for autonomous surface vehicles: preliminary results for marine litter detection**
Mellone, G., De Vita, C. G., Sánchez-Gallegos, D. D., Di Luccio, D., Mattei, G., Peluso, F., ... & Montella, R
31st Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP) (pp. 169-172). IEEE.

2023 **Blockchain-based schemes for continuous verifiability and traceability of IoT data.**
Rendon, C. M., González-Compeán, J. L., Sánchez-Gallegos, D. D., & Carretero, J.
31st Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP) (pp. 169-172). IEEE.

2022 **SecMesh: An efficient information security method for stream processing in edge-fog-cloud**
Torres-Charles, C. A., Carrizales-Espinoza, D. E., Sanchez-Gallegos, D. D., Gonzalez-Compean, J. L., Morales-Sandoval, M., & Carretero, J.
In Proceedings of the 2022 7th International Conference on Cloud Computing and Internet of Things (pp. 8-16).

2022 **On the building of self-adaptable systems to efficiently manage medical data**
Sanchez-Gallegos, G., Sánchez-Gallegos, D. D., Gonzalez-Compean, J. L., & Carretero, J.
In 2022 22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid) (pp. 985-992). IEEE.

2022 **SeRSS: a storage mesh architecture to build serverless reliable storage services**
Carrizales-Espinoza, D., Sánchez-Gallegos, D. D., Gonzalez-Compean, J. L., Carretero, J., & Marcelin-Jimenez, R.
In 2022 30th Euromicro International Conference on Parallel, Distributed and Network-based Processing (PDP) (pp. 88-91). IEEE.

2021 **A federated content distribution system to build health data synchronization services**
Carrizales-Espinoza, D., Sánchez-Gallegos, D. D., Gonzalez-Compean, J. L., & Carretero, J.
In 2021 29th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP) (pp. 1-8). IEEE.

2019 **A microservice-based building block approach for scientific workflow engines: Processing large data volumes with DagOnStar**
Sánchez-Gallegos, D. D., Di Luccio, D., Gonzalez-Compean, J. L., & Montella, R.
In 2019 15th International Conference on Signal-Image Technology & Internet-Based Systems (SITIS) (pp. 368-375). IEEE.

2019 **Internet of things orchestration using dagon workflow engine**
Sánchez-Gallegos, D. D., Di Luccio, D., Gonzalez-Compean, J. L., & Montella, R.
In 2019 IEEE 5th World Forum on Internet of Things (WF-IoT) (pp. 95-100). IEEE.

2019 **A data preparation approach for cloud storage based on containerized parallel patterns**
Carrizales, D., Sánchez-Gallegos, D. D., Reyes, H., González-Compeán, J. L., Morales-Sandoval, M., Carretero, J., & Galaviz-Mosqueda, A.
In Internet and Distributed Computing Systems: 12th International Conference, IDCS 2019, Naples, Italy, October 10–12, 2019, Proceedings 12 (pp. 478-490). Springer International Publishing.

2018 **A containerized service for clustering and categorization of weather records in the cloud**
Sánchez-Gallegos, D. D., Gonzalez-Compean, J. L., Alvarado-Barrientos, S., Sosa-Sosa, V. J., Tuxpan-Vargas, J., & Carretero, J.
In 2018 8th International Conference on Computer Science and Information Technology (CSIT) (pp. 26-31). IEEE.

2018 **An interoperable cloud-based geoportal for discovery and management of earth observation products**
Sánchez-Gallegos, D. D., Gonzalez-Compean, J. L., Sosa-Sosa, V. J., Marin-Castro, H. M., & Tuxpan-Vargas, J.
Comput. Sci. Inf. Technol.(CSIT).

Posters

2025 **AI and HPC for intense rain event early warning leveraging real-time weather radar**
Diana Di Luccio, Ciro Giuseppe De Vita, Gennaro Mellone, Dante D Sánchez-Gallegos, Pasquale Corvino, Mario Di Sarno, Pasquale De Luca, Emanuel Di Nardo, Vincenzo Capozzi, Vincenzo Bucciero, Raffaele Montella
2025 IEEE International Conference on eScience (eScience)

2025 **Hi-WeFAI: High-performance computing for Weather nowcasting with Federated Artificial Intelligence**
De Micco F., Formisano F., Di Luccio D., Di Nardo E., De Luca P., De Vita C. G., Mellone G., Sánchez-Gallegos D., Capozzi V., Ciaramella A., Corvino P., Pirozzi A., and Montella R.
3rd CINI International Summer School on High-Performance Computing

2025 **Xook-Flow: an abstract communication channel model for the data exchange in computing continuum**
Catherine Alessandra Torres Charles, Dante D Sanchez-Gallegos, Diana Carrizales-Espinoza, Jesus Carretero, Jose L Gonzalez-Compean
40th ACM/SIGAPP Symposium on Applied Computing

Chapter books

2022 **Muyal-Nez: servicios agnósticos para la creación de sistemas de ciencia de datos en e-salud**
Dante Domizzi Sánchez Gallegos, Jose Luis Gonzalez Compeán, Ricardo Marcelin Jimenez & Ricardo Landa Becerra
In BigData en Salud: Tecnologías Emergentes y Aplicaciones (pp. 65-80), ISBN: 9786078857258, Montiel & Soriano Editores, S.A. de C.V.

2022 **Plataforma Tecnólogica para la Gestión, Aseguramiento, Intercambio y Preservación de Grandes Volúmenes de Datos en Salud: Muyal-Ilal**
Jose Luis Gonzalez Compeán, Diana Elizabeth Carrizales Espinoza, Dante Domizzi Sánchez Gallegos, & Juan Armando Barron Lugo
In BigData en Salud: Tecnologías Emergentes y Aplicaciones (pp. 105-121), ISBN: 9786078857258, Montiel & Soriano Editores, S.A. de C.V.

2022 **Reflexiones sobre el almacenamiento digital de las organizaciones**
Ricardo Marcelin Jimenez, Hugo G. Reyes Anastacio, Dante Domizzi Sánchez Gallegos, & Jose Luis Gonzalez Compeán
In BigData en Salud: Tecnologías Emergentes y Aplicaciones (pp. 139-147), ISBN: 9786078857258, Montiel & Soriano Editores, S.A. de C.V.

2019 **Modelo de construcción de flujos de procesamiento agnósticos para gestionar ciclos de vida de productos de observación de la tierra**
Dante Domizzi Sánchez Gallegos, & Jose Luis Gonzalez Compeán
In Avances en Ciencias en Ingeniería y Tecnologías Computacionales (pp. 139-147), ISBN: 9786079023621, Cinvestav Unidad Tamaulipas

Presentations on conferences, congresses, and workshops

January 2026 **3rd International Workshop on Malleability Techniques Applications in High-Performance Computing**
On modeling knowledge graphs for representing and explaining wide-area distributed storage systems
Osaka, Japan

November 2025 **20th Workshop on Workflows in Support of Large-Scale Science**
DAGonStore: Reliable Data Management for Workflows on the Computing Continuum with DynoStore and DAGonStar
St. Louis, USA

June 2025 **Jornadas SARTECO 2025**
Sistema de almacenamiento distribuido a gran escala para el manejo de datos en ambientes heterogeneos
Sevilla, Spain

September 2024 **Keynote speaker at 2nd Workshop INSTIL - cItizeN Science engagemenT based on Ict soLutions**
Talk: Orchestrating large-scale services with DagOnStar for accelerating the construction of eScience systems
Osaka, Japan

August 2024 **PhD Symposium at Euro-PAR 2024**
Talk: On the building of computing continuum systems using serverless abstractions
Madrid, Spain

August 2024 **1st workshop about High-Performance e-Science**
Talk: GeoNimbus: A serverless framework to build earth observation and environmental services
Madrid, Spain

June 2024 **Jornadas SARTECO 2024**
Arquitectura para la orquestación de datos en el cómputo continuo
A Coruña, Spain

September 2023 **Jornadas SARTECO 2023**
Modelo de sincronización para la mitigación de cuellos de botella en flujos de datos
Ciudad Real, Spain

November 2022 **TopTamaulipas 2022**
Talk: Servicios de construcción de estructuras para el manejo del ciclo de vida de datos médicos
Cinvestav Tamaulipas, México

October 2022 **Taller en Ciencia de datos en salud, contaminantes y medio ambiente**
Talk: Muyal-Nez: Servicios para la construcción de sistemas de ciencia de datos no estructurados
Demo: Construye tus sistemas de ciencia de datos no estructurados (imágenes, ECGs y espirometrías) mediante la herramienta de diseño de Muyal-Nez
Mérida, Yucatán, México

September 2022 **Taller en Cómputo y almacenamiento en la nube para ciencia de datos**
Talk: Sistemas agnósticos para la creación de Sistemas de Ciencia de datos en eSalud
Universidad Autónoma Metropolitana Unidad Iztapalapa, Iztapalapa, Ciudad de México, México

August 2022 **Taller de Big Data en Salud**
Demo: Herramientas Zamná y Painal
Talk: Sistemas agnósticos para la creación de Sistemas de Ciencia de datos en e-Salud
Benemérita Universidad Autónoma de Puebla (BUAP), Facultad de ciencias de la computación, Puebla, Puebla, México

August 2022 **Taller en Telemedicina y ciencia de datos**
Talk: Muyal-Nez: Servicios para la construcción de sistemas de ciencia de datos no estructurados
CICESE Unidad Foránea Monterrey, Monterrey, Nuevo León, México

November 2021 **TopTamaulipas 2021**
Demo: Construcción y manejo de estructuras de procesamiento agnósticas de la infraestructura
Cinvestav Tamaulipas, México

March 2021 **2021 29th Euromicro International Conference on Parallel, Distributed and Network-Based Processing (PDP)**
Presentation: A Federated Content Distribution System to Build Health Data Synchronization Services
Valladolid, Spain

October 2019 **12th International Conference on Internet and Distributed Computing Systems (IDCS 2019)**
Presentation: A Data Preparation Approach for Cloud Storage Based on Containerized Parallel Patterns
Naples, Italy

October 2018 **2018 8th International Conference on Computer Science and Information Technology (CSIT)**
Presentation: A containerized service for clustering and categorization of weather records in the cloud
Amman, Jordan

May 2018 **7th International Conference on Information Technology Convergence and Services**
Presentation An interoperable cloud-based geoportal for discovery and management of earth observation products
Vienna, Austria

Additional courses and certificates

2025 **The 3rd CINI Summer School on High-Performance Computing, Naples, Italy** CINECA
35.5 hours.

2020 **Project: Writing a Research Paper** University of California, Irvine | Online
Coursera certificate
Duration: 30 hours.

2020 **Understanding Research Methods** University of London | Online
Coursera certificate
Duration: 10 hours.

2019 **TOEFL ITP** Universidad Autónoma de Tamaulipas
Score: 516 points.

2019 **Arquitectura de software (contenedores y microservicios)** Cinvestav Tamaulipas, México
Duration: 10 hours.

2019 **Writing in the Sciences** Stanford University | Online
Coursera certificate.
Duration: 5 hours.

2017 **Procesamiento de grandes cantidades de datos** Cinvestav Tamaulipas, México
Duration: 10 hours.

2016 **Intensive english program – level 4.** The Language Company, Edmond, Oklahoma
Duration: one month.

Awards

October 2019 **Best paper award "Alfredo Petrosino"**
12th International Conference on Internet and Distributed Computing Systems, Naples, Italy
A data preparation approach for cloud storage based on containerized parallel patterns

2016 **Winner of Cinvestav Tamcoder 2016**
State programming contest organized by the Cinvestav Tamaulipas